

MATH X314 - Assessment 4B Design and Development Record

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Project context and intent

I chose IPL cricket because it combines a familiar sporting context with rich match-level and ball-by-ball data. My aim was to move beyond simply showing scores and create a story about how modern T20 batting has become more aggressive. I framed the project around scoring evolution, batting behaviour, and team adaptation. When shaping the project, I prioritised audience clarity, visual consistency, and interpretation rather than technical complexity alone.

Changes from the proposal

The original proposal considered multiple cricket formats, but the final artefact focuses on the IPL. This change improved feasibility and consistency because the IPL dataset provided one coherent structure for all visualisations. Narrowing the scope allowed deeper exploration of scoring behaviour and clearer storytelling instead of spreading the analysis across incompatible data sources.

Data selection and preparation

The project uses Kaggle IPL match and delivery data. The matches file provides season, teams, winners, and match metadata, while the deliveries file provides ball-by-ball scoring information. I prepared innings totals by aggregating total runs by match, innings, and batting team. I also calculated batter-level runs, legal balls faced, strike rate, team win rate, and team-season average scores. A limitation is that the dataset represents IPL cricket only, so findings should not be generalised to all cricket formats.

Visualisation design decisions

Five visualisations were chosen to support different analytical tasks. A line chart shows the overall season trend because position along time is effective for communicating change. A boxplot shows medians, spreads, and outliers in first innings scores. A scatterplot shows the relationship between runs and strike rate, with transparency reducing overplotting. A horizontal bar chart ranks team win rates and improves readability of long team names. A heatmap combines team, season, and average score to reveal higher-order scoring patterns. I used a consistent blue-orange-red palette, minimal themes, ordered categories, and annotations to support clarity and reduce unnecessary visual noise.

Narrative and storytelling design

The story was structured as a progression from league-level change to player-level behaviour and then team-level success. The visualisations are ordered to first establish that scoring has risen, then show that high scores are more common, then explain how batters balance speed and volume, and finally connect this to team adaptation. Section transitions were added to improve continuity and guide the reader through the argument.

Technical implementation

The project was implemented using R/Quarto, tidyverse, ggplot2, scales, and plotly. The main technical challenge was reshaping ball-by-ball data into useful summary tables for innings, players, teams, and seasons. Another challenge was resolving column-name differences in the dataset, such as Season and batsman. These were handled by renaming columns consistently and documenting each transformation in the annotated code.

Reflection on project development

My understanding of the project changed from producing separate cricket charts to building a coherent data story. Feedback from earlier assessments showed that I needed stronger interpretation, consistent design, references, and better code annotation. In the final version, I focused more on visual hierarchy, transitions, and explaining why design choices supported the narrative. If repeating the project, I would explore more advanced player-level comparisons and possibly add a more detailed team-filtered dashboard.